

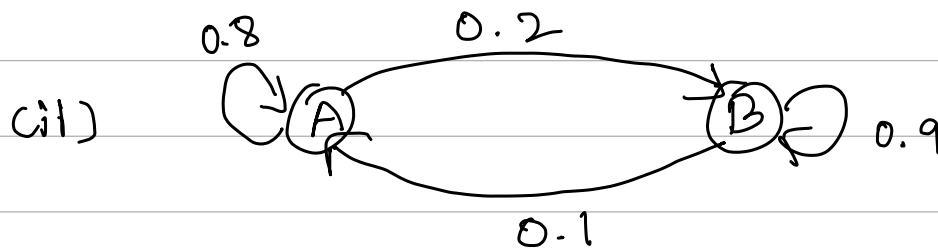
1) In a certain market, only 2 brands of cold drinks A and B are sold. Given that man last purchased brand A, there is 80% chance that he would buy some brand in the next purchase. While if the man purchased brand B, there is 90% chance that his next purchase would be brand B. Using this information,

(i) Develop TPM

(ii) Draw the transition diagram

(i) TPM is

$$\begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix} \end{matrix}$$



2) An urn initially contains 5 white balls and 5 black balls. The following experiment is repeated indefinitely. A ball is drawn from the urn. If the ball is white, it is put back in the urn. Otherwise, it is left out. Let x_n be the number of black balls remaining in the urn after n draws from the urn. Find TPM.

$$X_n = \{0, 1, 2, 3, 4, 5\}$$

There are always 5 white balls in the urn.

If there are i black balls in the urn, then total number of balls in the urn is $5+i$.

Probability of white ball drawn will be
$$\frac{5}{5+i}$$

Probability of black ball drawn will be
$$\frac{i}{5+i}$$

$P_{i,i}$ means process stays in the same state.
(a white ball is drawn and replaced)
 $P_{i,i-1}$ means a black ball is drawn and removed. The process moves to the next lower state.

$P_{i,i} \rightarrow i$ black ball & choose ball is white

$P_{i,i-1} \rightarrow i$ black ball & chosen ball is black

$\therefore P_{00} = \frac{5}{5+i} = 1$ (there are 0 black balls remaining i.e., all black balls are already removed)

$$P_{01} = P_{02} = P_{03} = P_{04} = P_{05} = 0$$

(there are 0 black balls are remaining therefore choosing black ball is impossible)

$$P_{10} = \frac{i}{5+i} = \frac{1}{5+1} = \frac{1}{6}$$

(one black ball remaining and black ball is chosen and removed)

$$P_{11} = \frac{5}{5+i} = \frac{5}{6} \quad (\text{chosen ball is white})$$

$$P_{12} = P_{13} = P_{14} = P_{15} = 0$$

(Only one black ball left, it can not be increase)

$$P_{20} = 0 \quad (2 \text{ black balls, at a time one ball is chosen})$$

$$P_{21} = \frac{2}{5+2} = \frac{2}{7} \quad (\text{chosen ball is black})$$

$$P_{22} = \frac{5}{5+2} = \frac{5}{7} \quad (\text{WB})$$

$$P_{23} = P_{24} = P_{25} = 0$$

$$P_{30} = 0, \quad P_{31} = 0$$

$$P_{32} = \frac{3}{5+3} = \frac{3}{8}$$

$$P_{33} = \frac{5}{5+3} = \frac{5}{8}$$

$$P_{34} = P_{35} = 0$$

$$P_{40} = 0$$

$$P_{41} = 0, \quad P_{42} = 0$$

$$P_{43} = \frac{4}{5+4} = \frac{4}{9}$$

$$P_{44} = \frac{5}{5+4} = \frac{5}{9}$$

$$P_{45} = 0$$

$$P_{50} = P_{51} = P_{52} = P_{53} = 0$$

$$P_{54} = \frac{5}{5+5} = \frac{5}{10}$$

$$P_{55} = \frac{5}{5+5} = \frac{5}{10}$$

$\therefore$ TPM is

	0	1	2	3	4	5
0	1	0	0	0	0	0
1	$\frac{1}{6}$	$\frac{5}{6}$	0	0	0	0
2	0	$\frac{2}{7}$	$\frac{5}{7}$	0	0	0
3	0	0	$\frac{3}{8}$	$\frac{5}{8}$	0	0
4	0	0	0	$\frac{4}{9}$	$\frac{5}{9}$	0
5	0	0	0	0	$\frac{5}{10}$	$\frac{5}{10}$

3) An urn initially contains 3 white balls and 2 black balls. The following experiment is repeated indefinitely. A ball is drawn from the urn. If the ball is white, it is put back in the urn. Otherwise, it is left out. Let X_n be the number of black balls remaining in the urn after n draws from the urn.

Find TPM.

$$S = \{0, 1, 2\}$$

White ball, then $P_{ii} = \frac{3}{3+i}$

Black ball, then $P_{i, i-1} = \frac{i}{3+i}$

$$P_{00} = \frac{3}{3} = 1$$

$$P_{01} = P_{02} = 0$$

$$P_{10} = \frac{1}{4} \quad (\text{Black})$$

$$P_{11} = \frac{3}{4} \quad (\text{White})$$

$$P_{12} = 0$$

$$P_{20} = 0$$

$$P_{21} = \frac{2}{5} \quad (\text{Black})$$

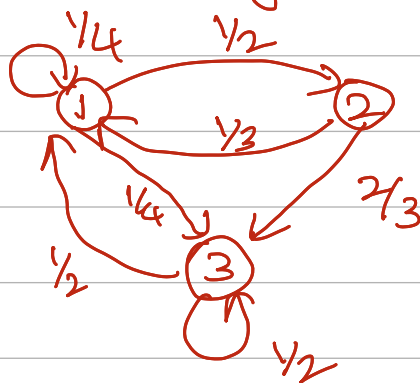
$$P_{22} = \frac{3}{5} \text{ (white)}$$

Transition matrix is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{4} & \frac{3}{4} & 0 \\ 0 & \frac{2}{5} & \frac{3}{5} \end{bmatrix} \end{matrix}$$

Higher Transition probability

1) Transition diagram is



(i) Find $P(X_4 = 3 \mid X_3 = 2)$

(ii) Find $P(X_3 = 1 \mid X_2 = 1)$

(iii) If we know $P(X_0 = 1) = 1/3$, find $P(X_0 = 1, X_1 = 2)$

(iv) If we know $P(X_0 = 1) = 1/3$, find $P(X_0 = 1, X_1 = 2, X_2 = 3)$

(v) Find TPM

$$(i) \quad P(X_4 = 3 \mid X_3 = 2) = P_{23} = 2/3$$

$$(ii) \quad P(X_3 = 1 \mid X_2 = 1) = P_{11} = 1/4$$

$$(iii) \quad \text{Given } P(X_0 = 1) = 1/3$$

$$\begin{aligned} P(X_0 = 1, X_1 = 2) &= P(X_0 = 1) P(X_1 = 2 \mid X_0 = 1) \\ &= 1/3 \times P_{12} \\ &= 1/3 \times 1/2 = 1/6 \end{aligned}$$

$$(iv) P(X_0=1, X_1=2, X_2=3)$$

$$= P(X_0=1) \times P(X_1=2 | X_0=1) \times P(X_2=3 | X_1=2)$$

$$= \frac{1}{3} \times P_{12} \times P_{23}$$

$$= \frac{1}{3} \times \frac{1}{2} \times \frac{2}{3} = \frac{1}{9}$$

(v) TPM is

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix} \end{matrix}$$

2) Let $\{X_n, n=0,1,2,\dots\}$ be a Markov chain with state space $\{0,1,2\}$ and the initial probability distribution $P(X_0=i) = \frac{1}{3}, i=0,1,2,\dots$

One step TPM is given by

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{bmatrix} \end{matrix}$$

Evaluate,

i) $P(X_1=1 | X_0=2)$

ii) $P(X_2=2 | X_1=1)$

iii) $P(X_2=2, X_1=1 | X_0=2)$

iv) $P(X_2=2, X_1=1, X_0=2)$

v) $P(X_3=1, X_2=2, X_1=1, X_0=2)$

Given $P(X_0=0) = P(X_0=1) = P(X_0=2) = 1/3$

(i) $P(X_1=1 | X_0=2) = P_{21} = \frac{3}{4}$

(ii) $P(X_2=2 | X_1=1) = P_{12} = \frac{1}{4}$

(iii) $P(X_2=2, X_1=1 | X_0=2)$

$$= \frac{P(X_2=2, X_1=1, X_0=2)}{P(X_0=2)}$$

$$= \frac{P(X_0=2) P(X_1=1 | X_0=2) P(X_2=2 | X_1=1)}{P(X_0=2)}$$

$$= \frac{\frac{1}{3} \times P_{21} \times P_{12}}{\frac{1}{3}} = \frac{\frac{1}{3} \times \frac{3}{4} \times \frac{1}{4}}{\frac{1}{3}} = \frac{3}{16}$$

(iv) $P(X_2=2, X_1=1, X_0=2)$

$$= P(X_0=2) P(X_1=1 | X_0=2) P(X_2=2 | X_1=1)$$

$$= \frac{1}{3} \times P_{21} \times P_{12} = \frac{1}{3} \times \frac{3}{4} \times \frac{1}{4} = \frac{1}{16}$$

(v) $P(X_3=1, X_2=2, X_1=1, X_0=2)$

$$= P(X_0=2) P(X_1=1 | X_0=2) P(X_2=2 | X_1=1) P(X_3=1 | X_2=2)$$

$$= \frac{1}{3} \times P_{21} \times P_{12} \times P_{12} = \frac{1}{3} \times \frac{3}{4} \times \frac{1}{4} \times \frac{3}{4} = \frac{3}{64}$$

Long run problems

1) Find the unique fixed probability vector t of

$$P = \begin{bmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 1/2 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Solⁿ:- Let $t = (x, y, z)$ be the fixed probability vector. Then $x + y + z = 1$

$$\Rightarrow z = 1 - x - y$$

Since t is prob vector,

$$tP = t$$

$$(x, y, 1-x-y) \begin{bmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 1/2 & 0 \\ 0 & 1 & 0 \end{bmatrix} = (x, y, 1-x-y)$$

$$\Rightarrow \frac{1}{2}y = x \Rightarrow y = 2x \Rightarrow 2x - y = 0 \quad \text{--- (1)}$$

$$\frac{3}{4}x + \frac{1}{2}y + 1 - x - y = y \Rightarrow -\frac{1}{4}x - \frac{3}{2}y = -1$$

$$\Rightarrow x + 6y = 4 \quad \text{--- (2)}$$

$$\frac{1}{4}x = 1 - x - y \Rightarrow \frac{5}{4}x + y = 1$$

$$\Rightarrow 5x + 4y = 4 \quad \text{--- (3)}$$